

Data Governance of a Semantic Data Layer

Scaling a scientific-intelligence knowledge graph into an enterprise semantic data layer
with FAIR-by-design services

Talk summary and knowledge-graph analysis, based on the speaker's slides

Speaker: Ben Gardner, AstraZeneca (Data Standards Interoperability & Governance, Data Office) – presented at
the Graphwise AI Summit

Overview

The talk describes how AstraZeneca moves from a **project-scale knowledge graph** (Scientific Intelligence) to an **enterprise semantic data layer**, and argues that the hard part is not the graph technology but **governance**: consistent identifiers, controlled vocabularies, data dictionaries, critical-data-element rules and a reference model, delivered as coordinated services in which FAIR is built in rather than bolted on.

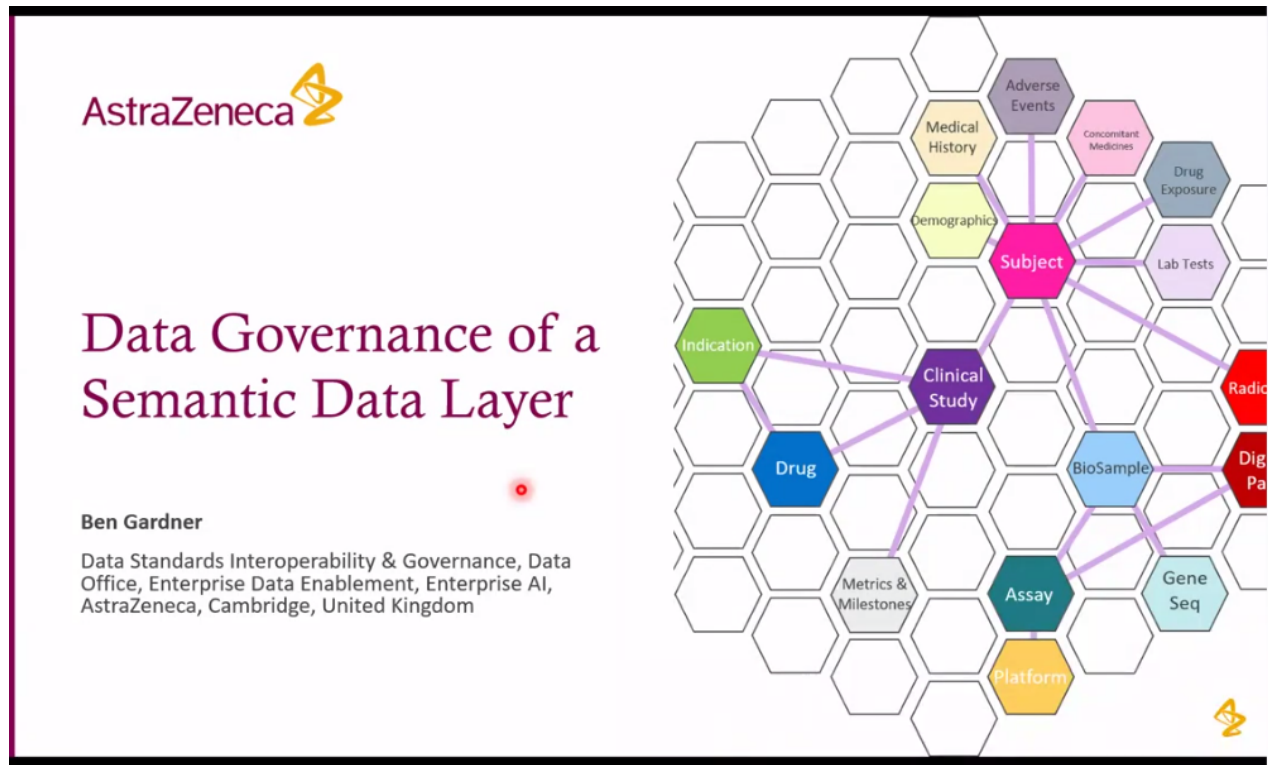
The document follows the speaker's storyline in five parts:

- **Part 1 – Context:** the Scientific Intelligence knowledge graph and the case for a semantic data layer (slides 1–4).
- **Part 2 – Governance framework:** information standards and FAIR-by-design services (slides 5–6).
- **Part 3 – The three FAIR services:** catalogue, reference/master data, data dictionaries (slides 7–10).
- **Part 4 – Trust and records:** critical data elements and the metadata records that implement them (slides 11–12).
- **Part 5 – Bringing it together:** AI-ready data, the unified knowledge graph and the semantic data layer (slides 13–16).

Core idea	Where it appears
Data product → semantic knowledge graph → semantic data layer	Slides 2, 3, 4
Governance expressed as a graph, using SKOS, ontology patterns and data contracts	Slide 5
FAIR services: Catalogue (F,A), Reference & master data (I), Data dictionaries (R)	Slides 6–10
Critical Data Elements (CDEs) link data fields to processes and rules	Slides 11–12
Unified knowledge graph serving curators, AI context and governance	Slides 13–16

Part 1 – Context: from a project graph to an enterprise layer

Title: Data Governance of a Semantic Data Layer

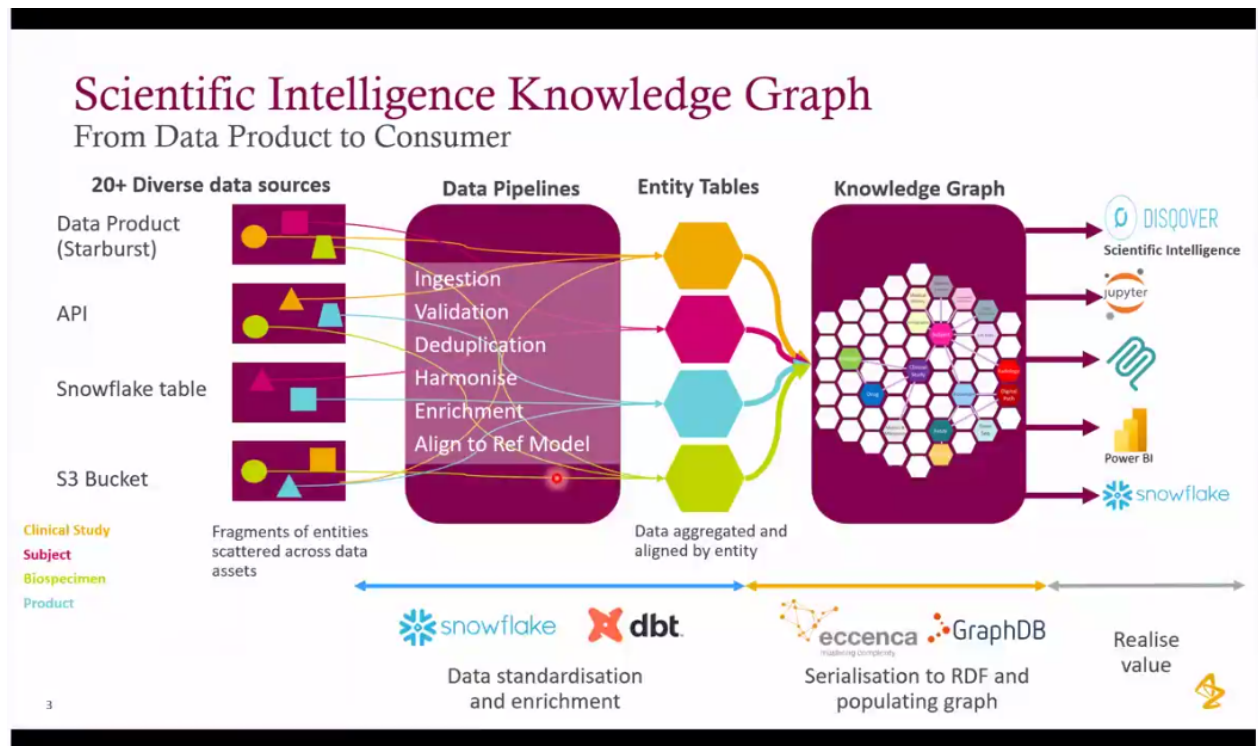


Slide 1: AstraZeneca – Ben Gardner, Data Standards Interoperability & Governance

The title slide frames the subject as *governance* of a semantic data layer, not its construction. The hexagon motif on the right is the visual language used throughout: each hexagon is an entity type from the reference model (Subject, Clinical Study, Drug, Indication, BioSample, Assay, Platform, Gene Seq, Lab Tests, Adverse Events, Medical History, Demographics, Drug Exposure, Concomitant Medicines, Metrics & Milestones and others), and the purple edges are the relationships between them.

Practitioner's note: this hexagon map is effectively a visual rendering of the enterprise reference model – a clinical-research domain ontology where Subject and Clinical Study act as hubs.

The Scientific Intelligence Knowledge Graph



Slide 2: From data product to consumer – pipeline from 20+ sources to the knowledge graph

More than twenty diverse sources (data products such as Starburst, APIs, Snowflake tables, S3 buckets) hold **fragments of entities scattered across data assets** – e.g. Clinical Study, Subject, Biospecimen and Product each appear partially in many places.

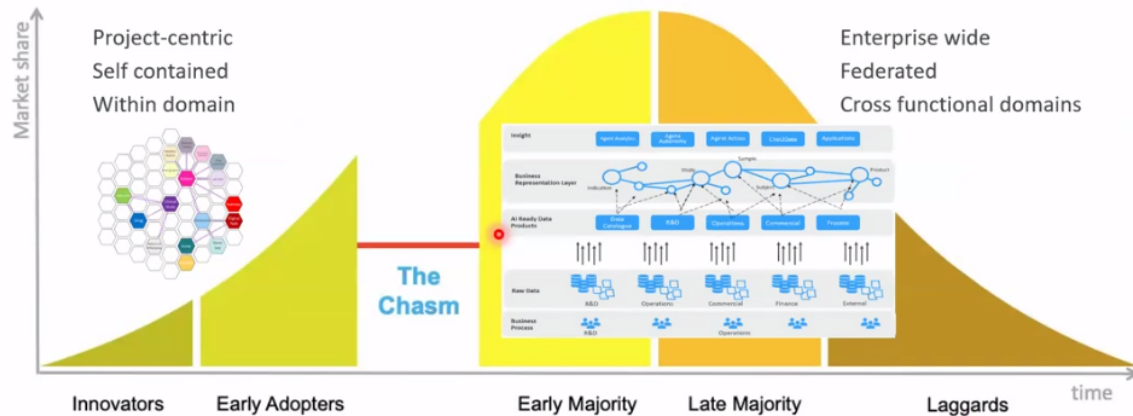
- **Data pipelines** perform ingestion, validation, deduplication, harmonisation, enrichment and alignment to the reference model.
- The output is a set of **entity tables** – data aggregated and aligned by entity.
- Entity tables are serialised to **RDF** and loaded into the **knowledge graph**.
- Tooling shown: Snowflake and dbt for data standardisation and enrichment; Eccenca and GraphDB for RDF serialisation and populating the graph.
- Consumers realise value through Discover (Scientific Intelligence), Jupyter, Power BI, Snowflake and other tools.

Practitioner's note: the pattern is a classic 'relational-to-RDF' lift: standardise in the warehouse first, then materialise entity-aligned tables as triples. Alignment to the reference model happens *before* serialisation, which keeps the graph consistent.

How to cross the chasm?

How to cross the chasm?

Scaling from a project knowledge graph to enterprise semantic data layer



Source: Moore "Crossing the Chasm"

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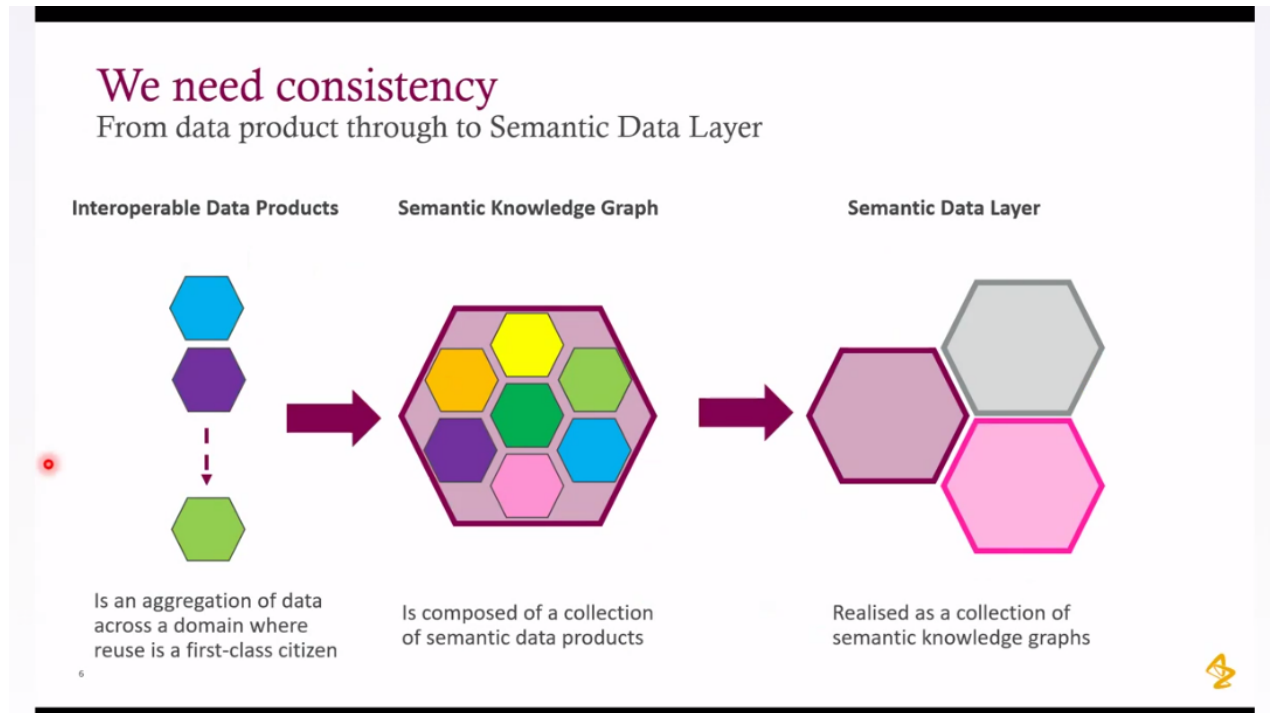
Slide 3: Scaling from a project knowledge graph to an enterprise semantic data layer (after Moore, 'Crossing the Chasm')

The technology-adoption curve is used as a metaphor. On the left, **innovators and early adopters** run project-centric, self-contained, within-domain graphs. On the right, the **early and late majority** need enterprise-wide, federated, cross-functional-domain capability. Between them lies **the chasm**.

The inset architecture shows the target: business processes and raw data in R&D, operations, commercial, finance and external areas; AI-ready data products above them; a business representation layer of connected concepts (indication, study, subject, sample, product); and insight on top (agent analytics, applications).

Practitioner's note: crossing the chasm is mainly an operating-model problem – shared identifiers, vocabularies and ownership – which sets up the rest of the talk.

We need consistency



Slide 4: From data product through to semantic data layer

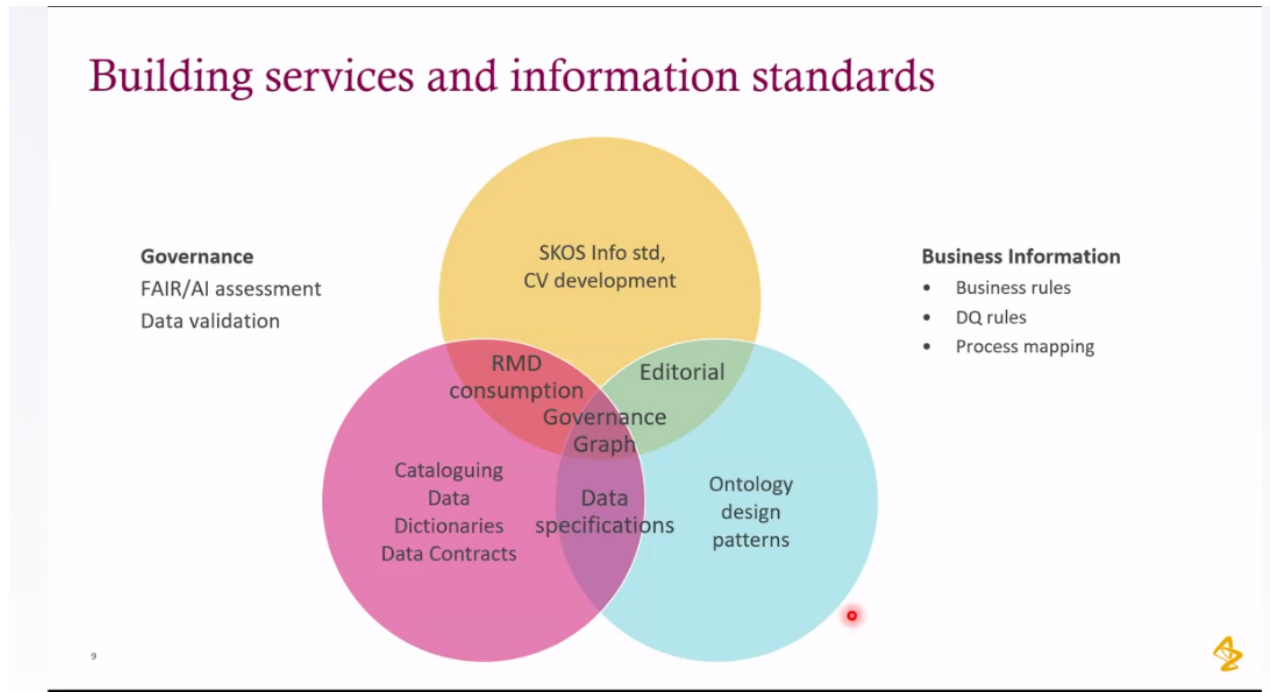
Three levels of aggregation are distinguished:

- **Interoperable data products** – an aggregation of data across a domain where reuse is a first-class citizen.
- **Semantic knowledge graph** – composed of a collection of semantic data products.
- **Semantic data layer** – realised as a collection of semantic knowledge graphs.

Practitioner's note: this is a federation model: the enterprise layer is not one monolithic graph but a set of governed graphs that interoperate because they share the same reference model and identifiers.

Part 2 – Governance framework

Building services and information standards



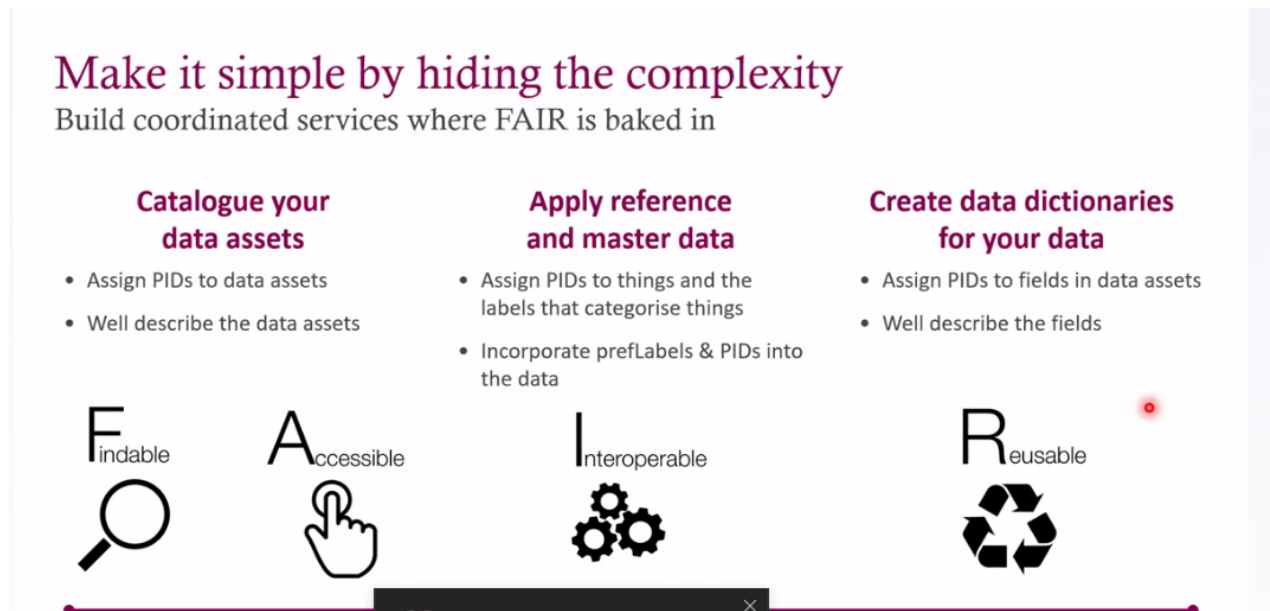
Slide 5: Governance, business information and the overlapping standards that make up a governance graph

Three overlapping disciplines are shown as a Venn diagram:

- **SKOS information standard and controlled-vocabulary development** (yellow).
- **Cataloguing, data dictionaries and data contracts** (pink).
- **Ontology design patterns** (blue).

Practitioner's note: the overlaps are the interesting part – reference/master-data (RMD) consumption sits where vocabularies meet cataloguing; editorial processes where vocabularies meet ontology patterns; **data specifications** where cataloguing meets ontology design; and the **governance graph** at the centre where all three converge. The side notes add governance activities (FAIR/AI assessment, data validation) and business information (business rules, data-quality rules, process mapping). Governance is therefore itself modelled as graph content.

Make it simple by hiding the complexity



Slide 6: Coordinated services where FAIR is baked in

The speaker maps FAIR to three coordinated services so that data owners are not asked to learn semantic-web theory:

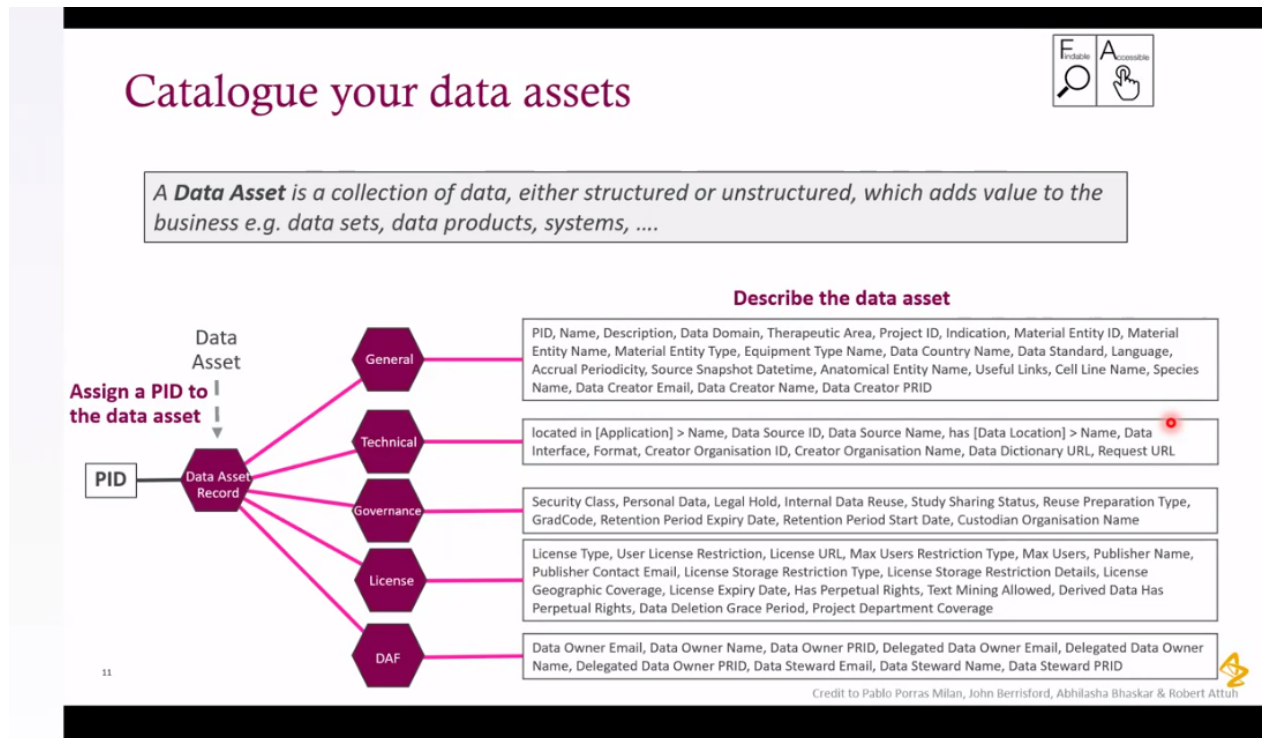
- **Catalogue your data assets** (Findable, Accessible): assign persistent identifiers (PIDs) to data assets and describe them well.
- **Apply reference and master data** (Interoperable): assign PIDs to things and to the labels that categorise them; incorporate prefLabels and PIDs into the data.
- **Create data dictionaries** (Reusable): assign PIDs to fields in data assets and describe the fields.

Practitioner's note: the key design choice is that PIDs are assigned at three levels – asset, thing/label and field – and all three live in one graph.

Part 3 – The three FAIR services

Service 1 – Catalogue your data assets (Findable, Accessible)

Catalogue your data assets



Slide 7: A data asset record with a PID, described across five facets

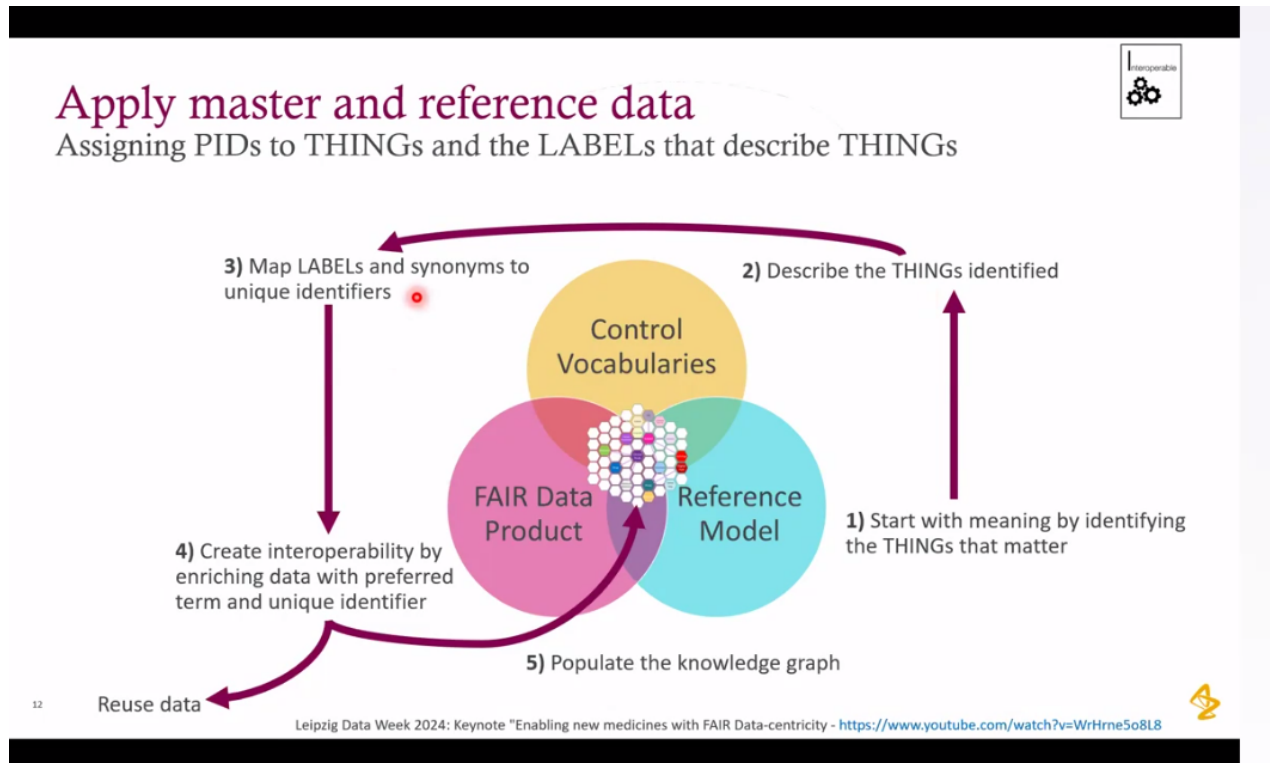
A **data asset** is a collection of data, structured or unstructured, that adds value to the business (data sets, data products, systems). Each asset receives a PID and a **Data Asset Record** described through five facets:

- **General** – name, description, data domain, therapeutic area, project, indication, material and equipment types, standard, language, periodicity, snapshot date, useful links, creator details.
- **Technical** – application and data source, data location, interface, format, creator organisation, data dictionary URL, request URL.
- **Governance** – security class, personal data, legal hold, internal reuse, study sharing status, reuse preparation, retention period start/expiry, custodian organisation.
- **License** – licence type, user and storage restrictions, licence URL, publisher, geographic coverage, expiry, perpetual rights, text-mining allowed, derived-data rights, deletion grace period.
- **DAF (data accountability)** – data owner, delegated owner, data steward, each with name, email and person ID.

Practitioner's note: separating facets keeps the record extensible and lets different teams own different parts (legal owns licence, security owns classification). Credit on slide: Pablo Porras Milan, John Berrisford, Abhilasha Bhaskar and Robert Attuh.

Service 2 – Apply reference and master data (Interoperable)

Apply master and reference data

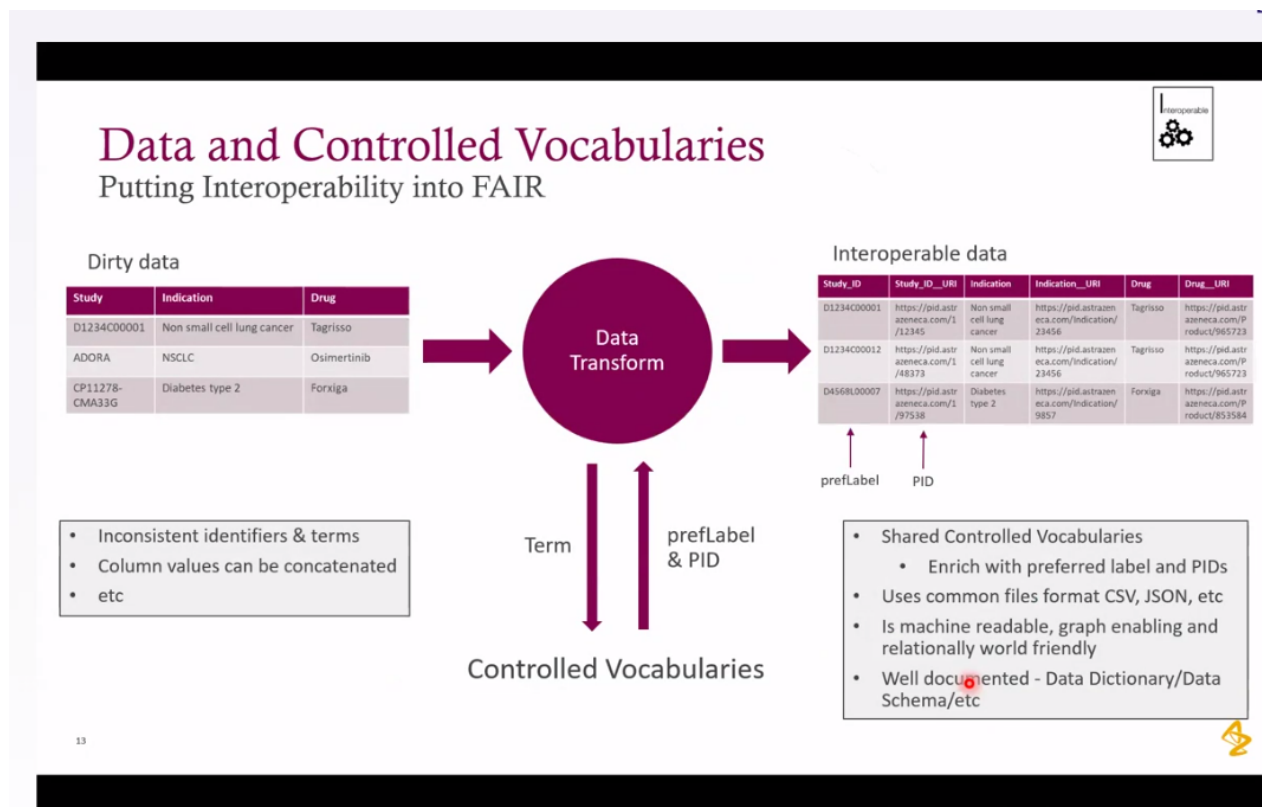


Slide 8: Assigning PIDs to THINGS and to the LABELS that describe THINGS

A five-step loop sits at the intersection of **controlled vocabularies**, the **reference model** and the **FAIR data product**:

- Start with meaning by identifying the THINGS that matter.
- Describe the things identified.
- Map LABELS and synonyms to unique identifiers.
- Create interoperability by enriching data with the preferred term and unique identifier.
- Populate the knowledge graph – and reuse the data.

Practitioner's note: separating THINGS (entities, with PIDs) from LABELS (terms and synonyms, also with PIDs) is the SKOS pattern: one concept, many labels, one prefLabel. Source: Leipzig Data Week 2024 keynote 'Enabling new medicines with FAIR data-centricity'.



Slide 9: Putting interoperability into FAIR – before and after

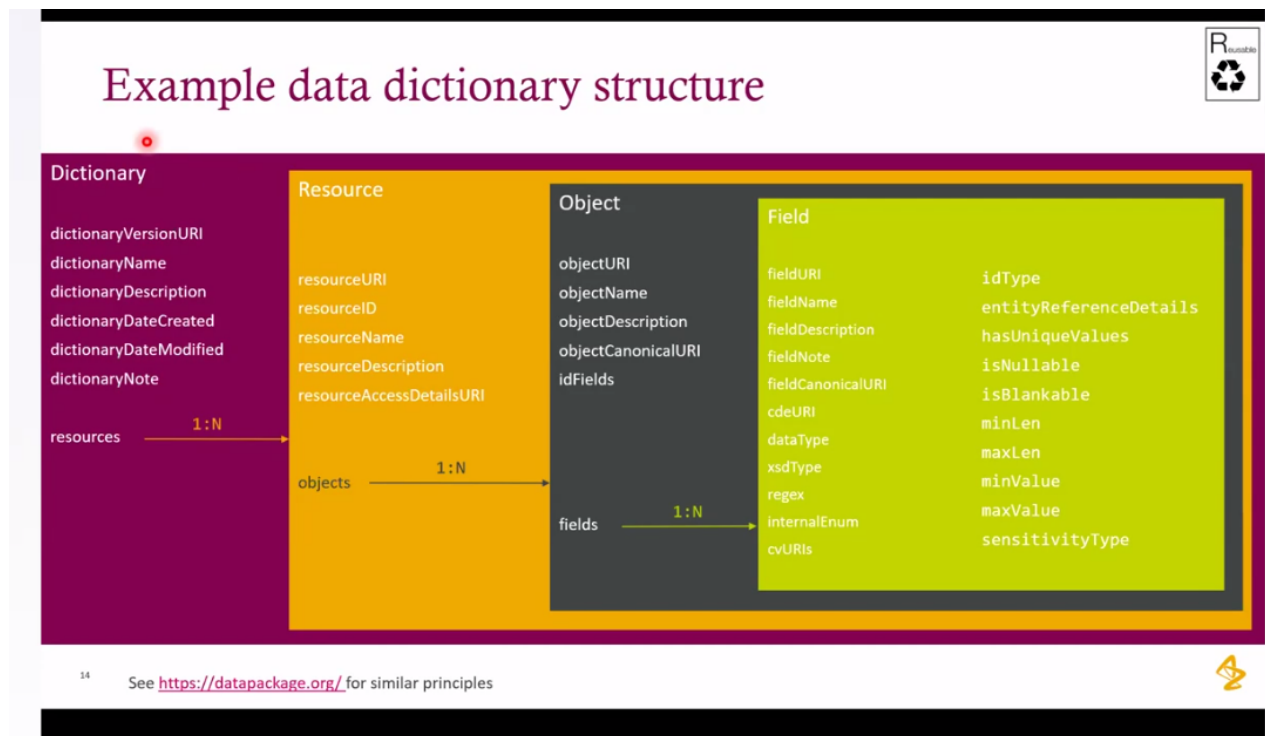
The slide shows the practical transformation. **Dirty data** has inconsistent identifiers and terms (for example 'Non small cell lung cancer' vs 'NSCLC', 'Tagrisso' vs 'Osimeritinib'), and values that may be concatenated. A **data transform** looks up each term in the controlled vocabularies and returns a **prefLabel** and **PID**, producing **interoperable data** with parallel columns such as Study_ID / Study_ID_URI, Indication / Indication_URI and Drug / Drug_URI.

- Shared controlled vocabularies enrich data with preferred labels and PIDs.
- Common file formats (CSV, JSON) are kept – the data is machine-readable, graph-enabling and still friendly to relational tools.
- Everything is documented in a data dictionary or schema.

Practitioner's note: adding a **_URI** column next to each label is a pragmatic way to make tabular data graph-ready without changing tools. Rows for the same drug resolve to the same PID, which is what enables joins across studies.

Service 3 – Create data dictionaries (Reusable)

Example data dictionary structure



Slide 10: Dictionary → Resource → Object → Field (each 1:N)

The dictionary is a four-level nested model:

- **Dictionary** – version URI, name, description, created/modified dates, note; contains *resources*.
- **Resource** – URI, ID, name, description, access-details URI; contains *objects*.
- **Object** – URI, name, description, canonical URI, ID fields; contains *fields*.
- **Field** – field URI, name, description, note, canonical URI, CDE URI, data type, XSD type, regex, internal enumeration, controlled-vocabulary URIs, ID type, entity-reference details, uniqueness, nullability, blankability, min/max length and value, sensitivity type.

Practitioner's note: note the **cdeURI** and **cvURIs** on the field – that is where the dictionary connects to critical data elements and to vocabularies. The slide points to datapackage.org for similar principles.

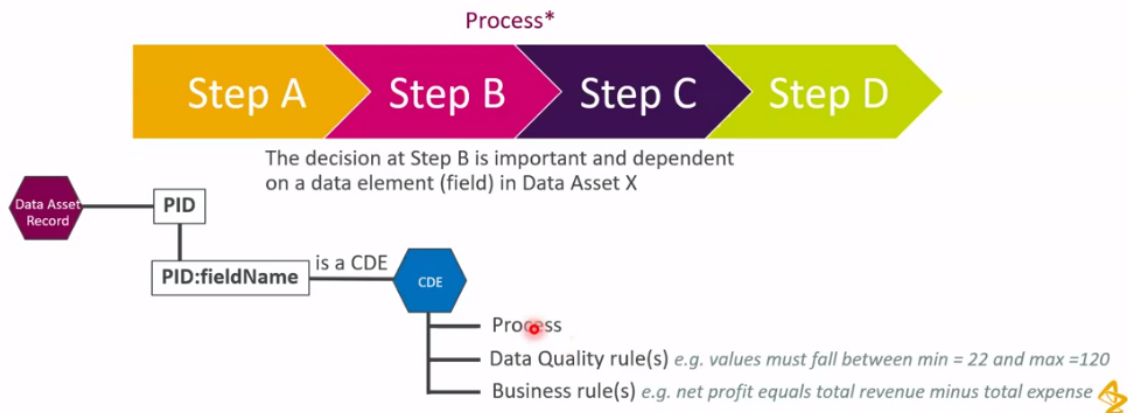
Part 4 – Trust: critical data elements and records

Data Governance

Data Governance

Extending to include TRUST categorisation - CDE

A **Critical Data Element** (CDE) is a Data Element (field) within a process that is determined to be vital to the successful operation of the organisation.

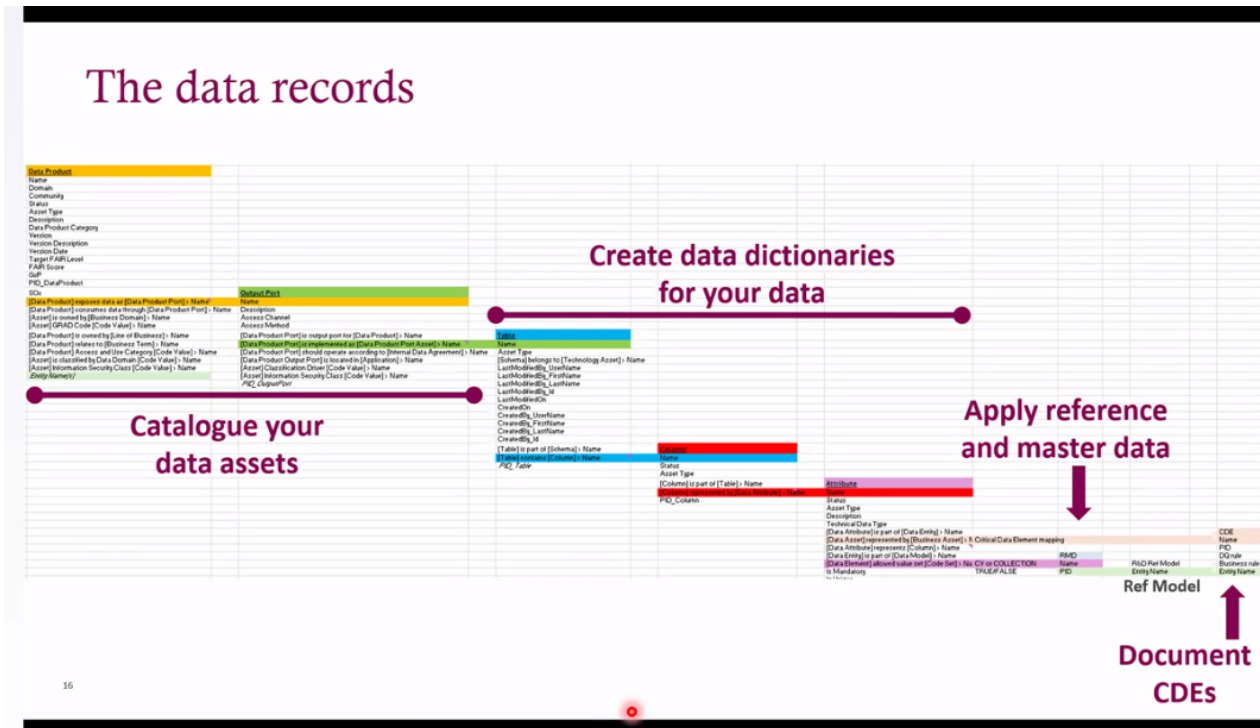


Slide 11: Extending to include TRUST categorisation – Critical Data Elements (CDE)

A **Critical Data Element** is a data element (field) within a process that is determined to be vital to the successful operation of the organisation. In the example, a process has Steps A to D; the decision at Step B depends on a field in Data Asset X. The field (addressed as PID:fieldName) *is a CDE*, and the CDE node links to its **process**, its **data-quality rules** (e.g. values must fall between a minimum of 22 and maximum of 120) and its **business rules** (e.g. net profit equals total revenue minus total expense).

Practitioner's note: this turns governance into graph traversal: from an asset record, to a field, to a CDE, to the process and rules that make it critical. Credit on slide: Di Elvers.

The data records



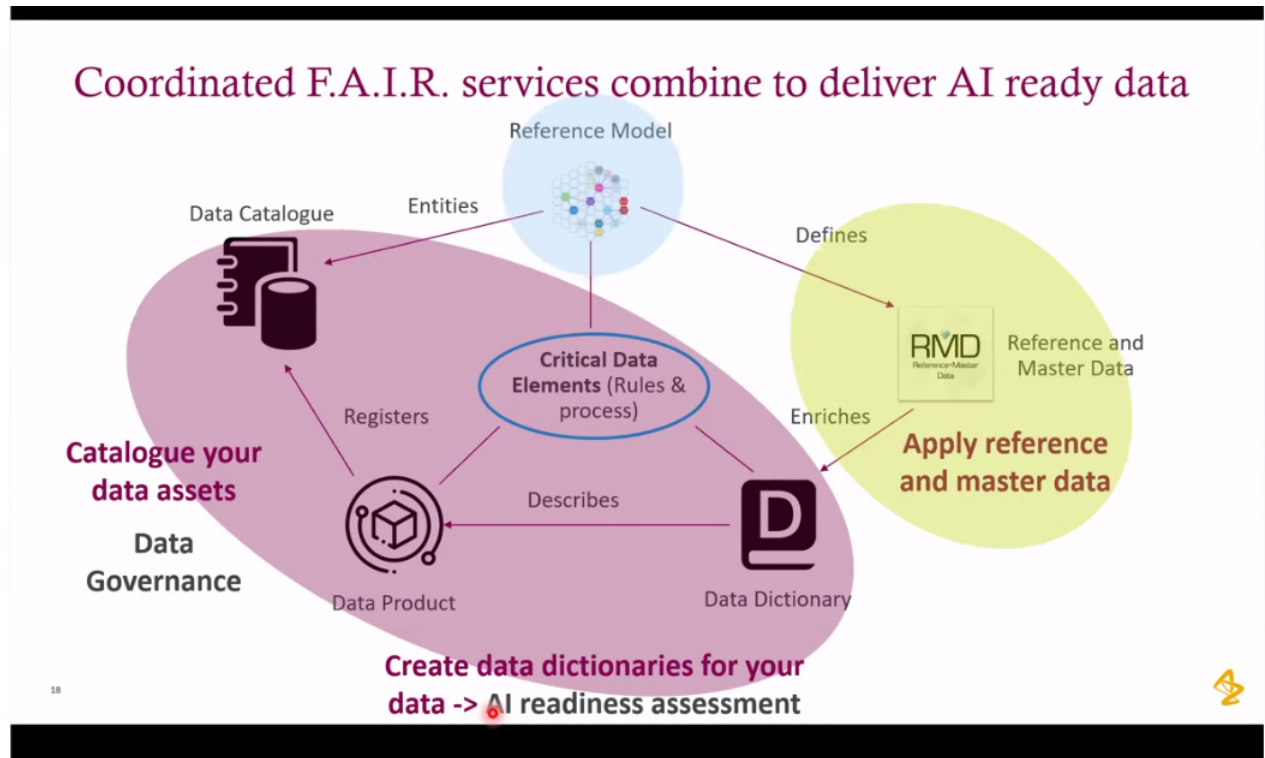
Slide 12: One spreadsheet-based record model covering catalogue, dictionary, reference data and CDEs

The slide shows the underlying record template used to capture everything above. Sections are annotated: **catalogue your data assets** (data product, output port, tables), **create data dictionaries** (table, column, attribute), **apply reference and master data** (reference-model entity, controlled-vocabulary/collection, PID) and **document CDEs** (CDE name, process, data-quality rule, business rule).

Practitioner's note: capturing all four service outputs in one structured template lets curators work in familiar tools while the content is loaded into the graph – a realistic on-ramp for governance teams. The text on this slide is small; treat it as an illustration of structure rather than a field list.

Part 5 – Bringing it together: AI-ready data and the semantic data layer

Coordinated F.A.I.R. services combine to deliver AI-ready data



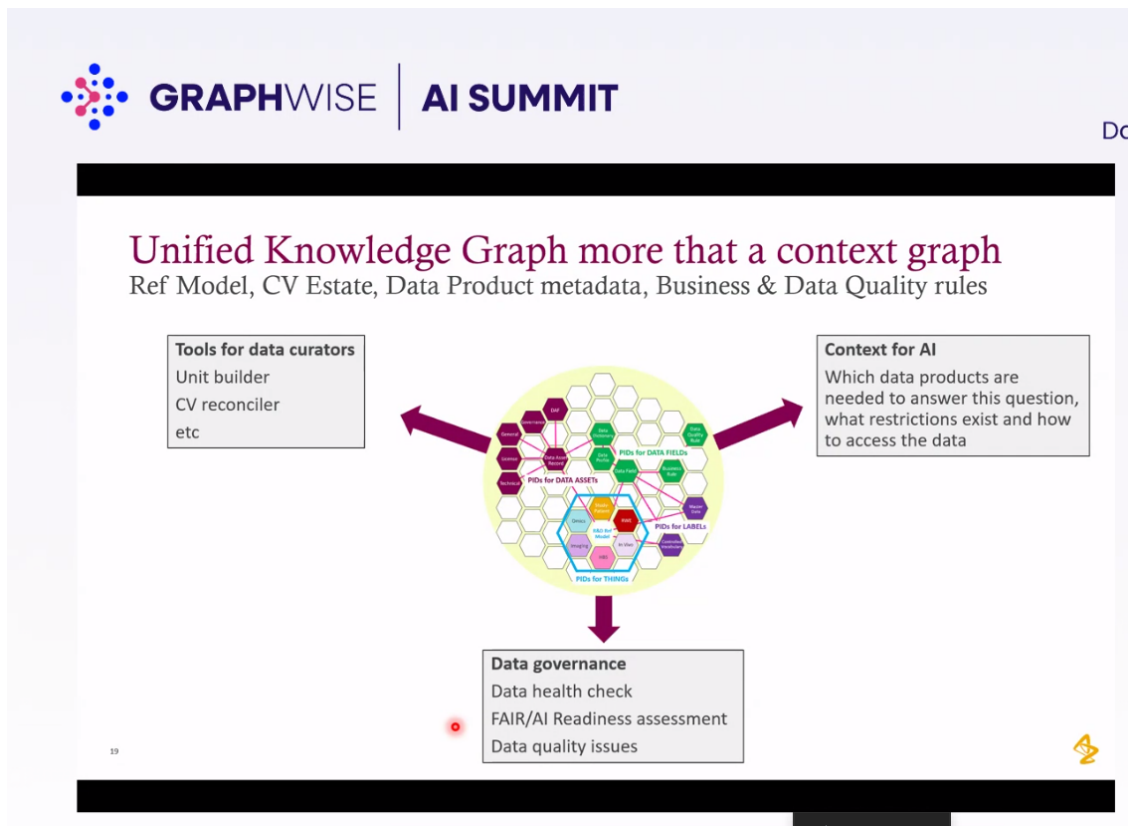
Slide 13: How the reference model, catalogue, dictionaries and reference/master data interlock

The **reference model** defines entities for the data catalogue and defines the **reference and master data (RMD)**, which in turn enriches the data dictionary. The **data dictionary describes the data product**, the data product **registers** in the catalogue, and **critical data elements** (rules and process) tie these together at the centre.

- Catalogue your data assets – this is the **data governance** core.
- Apply reference and master data.
- Create data dictionaries for your data – which supports an **AI readiness assessment**.

Practitioner's note: the AI-readiness assessment is a by-product: if assets are catalogued, fields are described and CDEs are governed, readiness can be queried rather than audited manually.

Unified Knowledge Graph: more than a context graph



Slide 14: Reference model, CV estate, data product metadata, business and data-quality rules

One graph combines PIDs for **things** (reference model), **labels** (controlled vocabularies), **data fields** (dictionary entries, business and data-quality rules) and **data assets** (asset records with general, technical, governance, licence and DAF facets). It serves three audiences:

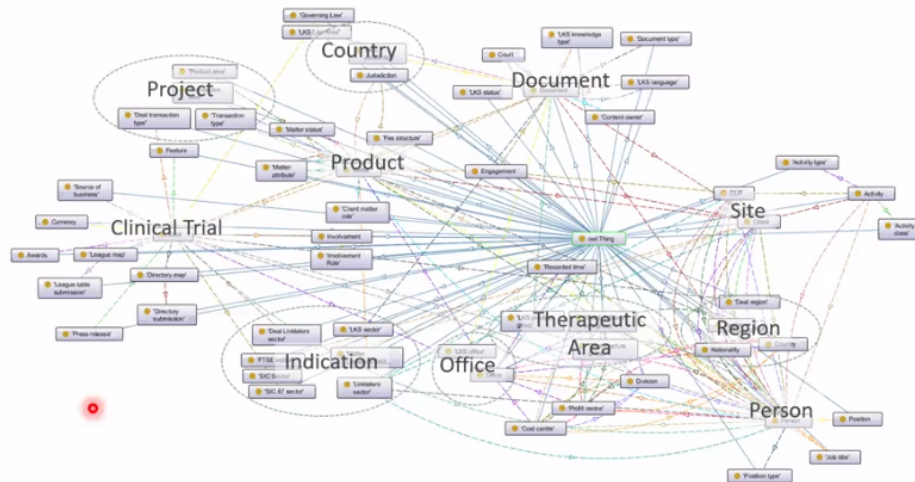
- **Tools for data curators** – e.g. a unit builder and a CV reconciler.
- **Context for AI** – which data products are needed to answer this question, what restrictions exist and how to access the data.
- **Data governance** – data health check, FAIR/AI readiness assessment, data-quality issues.

Practitioner's note: 'more than a context graph' means the graph holds not just meaning but also access, licence, quality and ownership – so an AI agent can decide not only *what* data answers a question but whether it is *permitted* and *trustworthy*.

What can we do with a Unified Knowledge Graph?

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What is the fragmentation state of entities across our data products? FAIR/AI Readiness, etc



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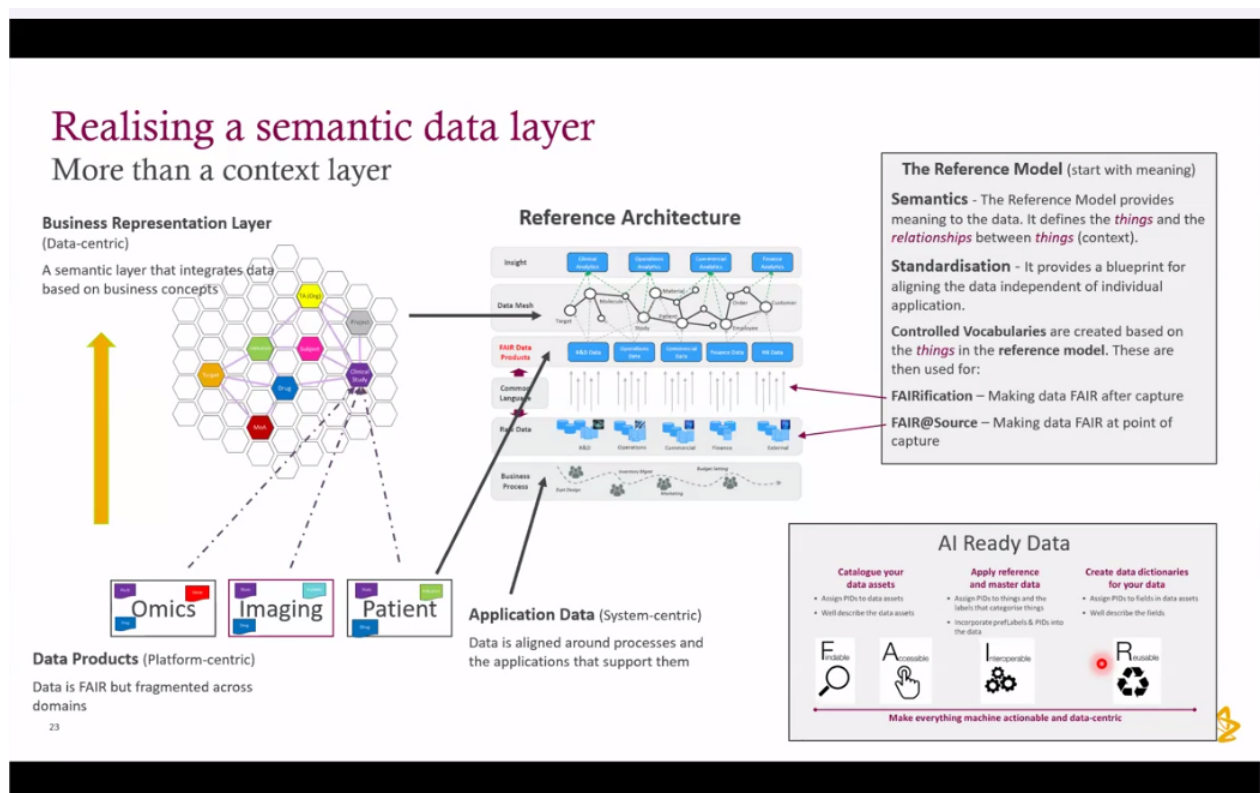
Image is illustrative not actual data

Slide 15: Measuring the fragmentation state of entities across data products (illustrative, not actual data)

The illustration shows entity types – Clinical Trial, Indication, Therapeutic Area, Product, Project, Country, Document, Site, Region, Office, Person – linked by many relationships. The question posed is: **what is the fragmentation state of entities across our data products?** With every asset, field and vocabulary term identified in one graph, the organisation can query where each entity type appears, how consistently, and how ready it is for FAIR and AI use.

Practitioner's note: the slide is explicitly labelled illustrative. The value claimed is analytic: fragmentation and readiness become measurable graph queries rather than anecdotes.

Realising a semantic data layer



Slide 16: More than a context layer – business representation layer over FAIR data products

Three layers are contrasted. **Application data** is system-centric (aligned to processes and the applications that support them). **Data products** are platform-centric (FAIR, but fragmented across domains – e.g. omics, imaging, patient). The **business representation layer** is data-centric: a semantic layer that integrates data based on business concepts, shown above the reference architecture (raw data → common language → FAIR data products → data mesh → insight).

- **Semantics** – the reference model gives meaning: it defines the things and the relationships between them.
- **Standardisation** – a blueprint for aligning data independent of any one application.
- **Controlled vocabularies** are created from the things in the reference model and used for **FAIRification** (making data FAIR after capture) and **FAIR@Source** (making data FAIR at the point of capture).
- **AI-ready data** – catalogue, apply reference/master data, create dictionaries; make everything machine-actionable and data-centric.

Practitioner's note: 'start with meaning' is the through-line of the whole talk: model the things first, derive vocabularies from them, then let catalogue, dictionary and CDE services reference those identifiers.

Key takeaways

- **Governance is the scaling constraint.** Crossing from a project graph to an enterprise layer depends on shared identifiers, vocabularies and ownership more than on tooling.
- **Three PID levels in one graph:** data assets, things/labels and data fields – all linked to the reference model.
- **FAIR as services.** Catalogue (F, A), reference and master data (I) and data dictionaries (R) hide the semantic complexity from data owners.
- **CDEs add trust.** Linking fields to processes, data-quality rules and business rules makes criticality queryable.
- **Federate, don't centralise.** The enterprise layer is a collection of semantic knowledge graphs built on one reference model.
- **AI readiness falls out of good metadata.** A unified graph tells AI which data products answer a question, what restrictions apply and how to reach the data.

All content is drawn from the speaker's slides and interpreted from a knowledge-graph perspective; detailed claims about AstraZeneca's internal implementation go no further than what the slides state.